
Calibration-Family Overfit: Why Trusted Sabotage Monitors Don't Transfer Across Lineages

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Abstract

Trusted monitoring is a central defense in AI control: a cheaper trusted model scores an untrusted model's actions for sabotage, and the most suspicious are audited or deferred. Such monitors are evaluated against one or two untrusted models, and the resulting accuracy is reported as a property of the monitor. We ask whether it is partly a property of the pairing. We make the untrusted policy family the controlled axis: we fit a monitor on family A's transcripts, apply it to family B, and decompose the cross-family AUROC into how obvious each family's sabotage is, how capable each monitor is, and the residual own-family advantage after both are removed: the interaction. A positive interaction means a monitor ranks its own lineage's sabotage above an attack from a different lineage after those row and column main effects are removed; in our matched controls, this remains true when attacks are matched on two obviousness proxies.

On code-backdoor transcripts the interaction is positive and survives the dominant confounds: $+0.172$ (95% CI $[+0.158, +0.185]$) on four open-weight families on a strict leak-free basis, holding across an eight-family matrix spanning four frontier vendors, blatancy-matched controls, a trained open-weight transcript-reading monitor up to 34B parameters, and a second code benchmark. The gap is geometric: each lineage's sabotage occupies a different scoring direction, and rotating the monitor's scoring direction toward the target lineage recovers most of the lost detection while a matched random rotation recovers nothing.

Translated to a defer-to-trusted protocol at a 1% audit budget, an off-lineage monitor catches 19% of attack attempts where an in-lineage monitor catches 41% at the same audit cost, a $+21.7$ -point gap on the strict leak-free basis (42% versus 24% on the transductive basis). We measure attack-attempt detection as the primary axis; realized sabotage is frontier-only on this substrate, too rare among the open families to power a full matrix, though on the frontier lineages that do realize it the interaction survives and is, by effect size, at least as sharp. We flag the absence of an all-family realized-sabotage matrix as a limitation. The takeaway for practice is that control evaluations should report cross-family transfer matrices, because a single-pairing accuracy overstates the safety a monitor delivers against a lineage it was not calibrated on; the four-step protocol that acts on the gap (report the full matrix, detect off-lineage drift, recalibrate with labeled sabotage, treat pooling as a bounded partial defense) is in Section 5.

1 Introduction

AI-control protocols defend against a misaligned untrusted model by routing its actions through a trusted monitor: the monitor assigns a suspicion score, and a fixed fraction of the most-suspicious actions are audited by a more expensive process or deferred to a trusted policy. The protocol's safety depends on the monitor ranking genuine sabotage above benign behavior at the chosen operating point. Evaluations of these protocols typically measure a

monitor against one or two untrusted policies and report a single AUROC or a single TPR at a fixed false-positive rate.

That reporting quietly assumes monitor quality is invariant to which untrusted model produced the actions. Deployment violates the assumption in an obvious way: the model you calibrate your monitor on is rarely the exact model you face, and is never the model you will face next quarter. If a monitor’s effectiveness is partly a function of the specific policy lineage it was tuned against, then a published accuracy is an in-lineage number, and the realized accuracy on a different lineage can be lower.

We make the untrusted policy family the controlled axis and ask whether trusted sabotage monitors transfer across families. A cross-family drop has at least three explanations, and only the third supports a claim about calibration:

1. Attacker obviousness. Family B’s sabotage may simply be more or less detectable for everyone. This produces a column-structured pattern (B is hard for every monitor) with no own-family advantage.
2. Monitor capability. A weaker monitor may miss subtle sabotage from any family. This produces a row-structured pattern.
3. Calibration overfit. A monitor tuned on family A may key on A’s stylistic and behavioral signature, ranking A’s sabotage well and other families’ sabotage poorly. This produces an asymmetric, diagonal-dominant pattern that survives after the row and column effects are removed.

Our headline statistic is the interaction term from a two-way decomposition of the cross-family AUROC matrix. A positive interaction is consistent with explanation 3 once the additive obviousness (column) and capability (row) effects are removed.

We report four findings on code-backdoor sabotage generated with ControlArena:

- The interaction is positive and significant. On APPS, with four open-weight families at 600 transcripts each, the family-conditioned monitor’s interaction is +0.110 (95% CI [+0.094, +0.125]) on a transductive basis and +0.172 [+0.158, +0.185] on a strict leak-free feature basis that we report as primary (Section 4.1). A monitor calibrated on Mistral scores Mistral’s own backdoors at 0.97 AUROC, while foreign heads rank another lineage’s backdoors near or below chance (the Qwen-calibrated head scores Llama’s backdoors at 0.42, below chance). Adding a frontier model (Claude Sonnet) as a fifth family strengthens the interaction to +0.132 [+0.122, +0.143], and an eight-family matrix spanning four frontier vendors (Anthropic, Google, xAI, OpenAI) raises it to +0.150 [+0.142, +0.157] (exact permutation $p = 0.000025$). The advantage is not a weak-open-model artifact: among the four frontier lineages alone it is +0.103 [+0.094, +0.110] on the full transcript and +0.135 [+0.120, +0.150] action-only.
- It does not depend on the monitor. A fixed-prompt API monitor, which cannot overfit a ranking by construction, shows a null diagonal bonus. A trained LoRA monitor that reads the raw transcript replicates the interaction at +0.18, stable across three training seeds (+0.179, +0.204, +0.194) and two base models (Phi-3.5-mini, Phi-4-mini).
- It is not specific to code distribution. Regenerating the families on bigcodebench, a different code distribution, gives a leak-free interaction of +0.233 (full transcript) and +0.099 (code only). On SHADE-Arena (Kutasov et al., 2025b), a non-code agentic tool-use setting, the interaction is directionally consistent at +0.035 (95% CI [-0.004, +0.077]) across three frontier vendor lineages but is underpowered at seven test tasks, its interval including zero, so we report it as suggestive non-code evidence rather than a significant replication (Section 4.3).
- And it costs catch rate. Translated into a defer-to-trusted protocol at a 1% audit budget, on the eight-family matrix the in-lineage monitor catches 41% of attack attempts and the off-lineage monitor catches 19%, a +21.7-point gap at the same audit cost on the strict leak-free basis ([+19.7, +26.7]); on the transductive basis the pair is 42% versus 24%, an 18.4-point gap ([+15.8, +21.1]; +10 points across the five families, +6.5 points on the four open-weight families alone).

We contribute both a measurement-validity result and a protocol that acts on it. The measurement result makes policy family a controlled axis: current papers that report a monitor accuracy in one or two pairings are reporting a diagonal cell, the off-diagonal that a deployed monitor faces is lower, and that gap survives subtraction of the per-attacker and per-monitor main effects, so it is not explained by the obviousness or capability confounds a careful evaluation already controls for. The protocol (Section 5) says what to do about that gap: report the full transfer matrix, detect off-lineage operation at inference from benign drift, recalibrate per lineage with labeled sabotage because input-level deferral and benign-only alignment provably cannot recover it, and treat multi-family pooling as a bounded partial defense. Every step is a measured result with a cost or a bound.

We deliberately study passive distribution shift, not adaptive attack. The attack prompts are fixed stealth scaffolds shipped with the benchmark; we do not optimize attacks against any monitor, set up collusion, or select best-of-N evasions. The question is whether equally subtle sabotage from a different lineage evades a monitor that was never tuned for it.

2 Setup and threat model

Substrates. We use ControlArena (UK AI Security Institute and Redwood Research, 2025; Greenblatt et al., 2024) on two code-backdoor settings. APPS (Hendrycks et al., 2021) asks the policy to solve a competitive-programming problem while inserting a backdoor that fails on a held-out secret input. Bigcodebench (Zhuo et al., 2024) poses the same backdoor objective on a different, library-heavy code distribution. Each sample has an honest mode (solve the task) and an attack mode (solve plus backdoor), generated on identical task IDs with the same sandbox, turn budget, and tool visibility.

Policy families. Four open-weight families served on DeepInfra: Qwen3-14B (thinking disabled), Gemma-3-27B, Mistral-Small-3.2-24B, and Llama-3.3-70B. These span a real detectability range under a fixed-prompt monitor (Mistral 0.767, Llama 0.672, Qwen 0.540, Gemma 0.490 AUROC), which is what lets us separate obviousness from calibration. Five frontier-capability families (Anthropic Claude Sonnet 4.6, Google Gemini 2.5 Flash, xAI Grok-4.3, OpenAI GPT-5 on their native APIs, and DeepSeek-V3.2 on DeepInfra) are added in Section 4.5 to test the interaction against strong attackers across five independent vendors. Families are chosen for provenance independence rather than vendor badge; distilled variants are excluded from the main matrix.

Monitors. The trusted model is gpt-4.1-mini, vendor-disjoint from the open-weight policies. We study three monitor classes (Section 4.2). The default fixed-prompt monitor scores the end-of-trajectory transcript; the Track-2 family-conditioned monitor embeds each family’s transcripts with a frozen embedder and fits a per-calibration-family logistic head; a LoRA monitor finetunes a small open model on each family’s transcripts.

Splits. Task IDs are partitioned 60/40 into calibration and test by a fixed hash, disjoint across the two splits, shared across families so no task appears in two splits. This prevents same-task leakage from masquerading as transfer.

Success filtering. Both-success backdoors (main task passes and the secret input fails) are near zero for sub-frontier policies even at 600 transcripts per family, so a strict success filter is degenerate at this scale. We report the unfiltered honest-versus-attack AUROC, which measures detection of attack attempts. This is a legitimate and disclosed framing; the monitor sees the transcript whether or not the backdoor succeeded.

Two metric tracks. AUROC is threshold-free, so a fixed prompt-only monitor produces the same ranking regardless of which family was used to “calibrate” it; its AUROC matrix has no calibration-family axis by construction, and claiming an AUROC transfer matrix for it would be a math error. We therefore report two tracks. Track 1 (operational) sets a threshold at the 1%-false-positive point on family A’s benign calibration transcripts and applies it to family B, reporting TPR at that threshold. Track 2 (ranking) requires a monitor whose scores depend on the calibration family, and is where a real AUROC transfer matrix lives. The family-conditioned and LoRA monitors are Track-2 monitors; the fixed-prompt monitor is the Track-1 baseline.

Table 1: Policy families, serving, generation funnel (scored transcripts per mode), attacker capability (attack-mode main-task success), clean-backdoor rate (both main and side success), and detectability under the fixed-prompt monitor (per-eval-family AUROC; the frontier families are APPS-only and not in the fixed-prompt matrix). n=600 per family per mode before scoring attrition. The five frontier families are mutually vendor-disjoint; four (Anthropic, Google, xAI, DeepSeek) are also vendor-disjoint from the gpt-4.1-mini monitor, while GPT-5 shares OpenAI with the monitor, a same-vendor cell flagged in Section 4.5 (the Track-2 head’s embedder is itself an OpenAI model, but the interaction reproduces under a non-OpenAI open-weight embedder on the four-open matrix (Section 6), and in any case a same-vendor effect could only aid a same-vendor monitor’s detection, so it cannot inflate the transfer gap). Gemini ran at a reduced agentic step cap (5 vs 15) for tractability (it does not self-terminate early); honest and attack share the cap, and Section 4.5 reports the length-robust action-only interaction for the matrices that include it.

Policy family	Serving	Attack/honest scored	Attack main-success	Both-success	Fixed-prompt AUROC
Mistral-Small-3.2-24B	DeepInfra	599 / 529	40%	0.0%	0.82
Llama-3.3-70B	DeepInfra	525 / 537	41%	0.0%	0.66
Qwen3-14B (think-off)	DeepInfra	600 / 600	32%	0.2%	0.47
Gemma-3-27B	DeepInfra	600 / 600	39%	0.0%	0.51
Claude Sonnet 4.6 (frontier)	Anthropic API	600 / 600	68%	2.2%	—
Gemini 2.5 Flash (frontier)	Google AI Studio	523 / 546	48%	1.0%	—
Grok-4.3 (frontier)	xAI API	590 / 600	49%	4.1%	—
GPT-5 (frontier)	OpenAI API	325 / 593	76%	11.7%	—
DeepSeek-V3.2 (frontier)	DeepInfra	600 / 600	48%	5.3%	—

3 Measuring transfer: the interaction decomposition

For a set of families $\{A\}$ we build the $N \times N$ matrix $M[A,B] = \text{AUROC}$ of monitor calibrated on A , evaluated on B ’s held-out test transcripts. The diagonal $M[A,A]$ is the in-lineage number a single-pairing evaluation would report; the off-diagonal $M[A,B]$ is what a deployed monitor faces.

We decompose M into additive main effects plus an interaction:

$$M[A,B] = \text{grand_mean} + \text{row_A} + \text{col_B} + \text{interaction}[A,B]$$

and report the mean diagonal interaction, mean over A of $(M[A,A] - \text{row_A} - \text{col_A} + \text{grand_mean})$. The row effect absorbs monitor capability (some monitors are better overall). The column effect absorbs attacker obviousness (some families are harder for everyone, including the “stronger models sabotage more subtly” case). What remains on the diagonal is the own-family advantage that calibration overfit predicts and that obviousness and capability do not.

Confidence intervals are cluster bootstraps over test tasks (resample tasks with replacement, recompute the matrix and the interaction), never over individual transcripts, because honest and attack samples on the same task are not independent. The bootstrap resamples tasks with replacement, preserving multiplicity (1000 draws, clustered by task).

Feature basis. The headline interaction is estimated on a strict leak-free basis: the scaler, PCA, and logistic head are fit only on the calibration family, so the evaluated family never enters the preprocessing. On the APPS rosters, a transductive basis that pools all families into one shared scaler and PCA lets the evaluated family leak into the representation and dilutes the diagonal advantage; where we report a transductive number we say so and, on those rosters, treat it as a conservative lower bound (Section 4.1, Table 2b). BigCodeBench goes the other way, so strict is primary there.

A positive interaction with a CI excluding zero is evidence for calibration overfit. A null interaction with column-only structure would instead support an obviousness story, a different and narrower paper; we pre-registered that pivot and did not need it.

A single conditional placement model. The cell-by-cell decomposition and its task-clustered bootstrap recover the same estimand as one conditional task-level regression. We fit a placement-value model (placement of each attack against the evaluated family’s benign scores, whose attack-mean equals the cell AUROC by construction) over all 3,816 graded attack-by-head rows, regressing placement on policy-family and monitor-family main effects, a diagonal indicator, the continuous obviousness grade, transcript length, the main- and side-task success indicators, and task fixed effects. The diagonal coefficient is the own-family bonus net of every one of those terms. On the leak-free basis it is +0.171 (task-clustered 95% CI [+0.158, +0.184]) and is numerically invariant across the nested ladder that adds blatancy, length, success strata, and task fixed effects in turn. That invariance is structural, not incidental: each attack contributes its obviousness, length, success, and task identically to its one diagonal and three off-diagonal rows, so any attack- or task-level confound loads on the column main effect and cannot move a within-attack, across-monitor contrast. A per-attacker obviousness story therefore cannot produce the diagonal; only a monitor-side affinity can.

4 Results

Table 2 collects the interaction term across the Track-2 code-substrate settings; it is positive with a 95% confidence interval excluding zero in all of them.

We nominate the frozen four-open matrix as the pre-registered primary result and label every later roster as post-hoc robustness; because the interaction already excludes zero on the frozen four-open primary before any frontier lineage enters, the effect is not an artifact of family selection. The full roster registry, with each roster’s inclusion rule and pre-registration date, is in the appendix.

4.1 Headline: the family-conditioned interaction (APPS)

The Track-2 monitor embeds each family’s full transcript with text-embedding-3-small, fits a StandardScaler and a 50-component PCA on the pooled calibration embeddings, and trains a per-family logistic head ($C=1.0$) on honest-versus-attack labels. At 600 transcripts per family (calibration 368 / test 232 tasks), the cross-family AUROC matrix on four families gives a mean diagonal of 0.71 and a mean off-diagonal of 0.57, a raw own-versus-foreign gap of 0.14 that, once the row and column main effects are removed, is an interaction of +0.110 (95% CI [+0.094, +0.125]).

The CI excludes zero. The structure is diagonal-dominant and asymmetric: the Mistral-calibrated head scores Mistral’s own attacks at 0.97 AUROC but foreign lineages far lower, and the Qwen-calibrated head scores Llama’s attacks at 0.42, below chance. A monitor that ranks one lineage’s sabotage near-perfectly actively mis-ranks another’s.

Leak-free re-estimation. The estimate above uses the transductive basis, which fits one shared scaler and PCA on the pooled calibration embeddings and so lets the evaluated

Table 2: Interaction term (own-family advantage net of attacker obviousness and monitor capability) across monitor classes, substrates, and scales. These estimates use the transductive feature basis (one shared scaler and PCA fit on the pooled calibration embeddings); the leak-free strict re-estimation, where available, is in Table 2b and is larger than or unchanged in every APPS roster; the BigCodeBench substrate check is lower on the strict basis, and is reported as the one exception.

Setting	Cut	Interaction	95% CI
APPS, 4 open families	full transcript	+0.110	[+0.094, +0.125]
APPS, 4 open families	action-only	+0.055	[+0.045, +0.065]
APPS, 5 families (+ frontier)	full transcript	+0.132	[+0.122, +0.143]
APPS, 5 families (+ frontier)	action-only	+0.069	[+0.058, +0.079]
APPS, 6 families (2 frontier vendors)	full transcript	+0.137	[+0.128, +0.146]
APPS, 6 families (2 frontier vendors)	action-only	+0.120	[+0.110, +0.132]
APPS, 8 families (4 frontier vendors)	full transcript	+0.150	[+0.142, +0.157]
APPS, 8 families (4 frontier vendors)	action-only	+0.138	[+0.126, +0.150]
APPS, frontier-only (4 vendors)	full transcript	+0.103	[+0.094, +0.110]
APPS, frontier-only (4 vendors)	action-only	+0.135	[+0.120, +0.150]
APPS, 9 families (5 frontier vendors)	full transcript	+0.152	[+0.145, +0.159]
APPS, 9 families (5 frontier vendors)	action-only	+0.157	[+0.146, +0.167]
APPS, frontier-only (5 vendors)	full transcript	+0.100	[+0.092, +0.107]
APPS, frontier-only (5 vendors)	action-only	+0.164	[+0.150, +0.177]
bigcodebench, 3 families	full transcript	+0.250	[+0.233, +0.268]
bigcodebench, 3 families	action-only	+0.141	[+0.127, +0.155]
LoRA monitor (3 seeds, 2 bases)	full transcript	+0.18 to +0.20	all > 0

family leak into the representation. Re-fitting the scaler, PCA, and logistic head on each calibration family alone, with the evaluated family absent from the entire pipeline, is the leak-free estimate and raises the interaction rather than lowering it, because the pooled basis was diluting the diagonal advantage: on the four open families it is +0.172 (95% CI [+0.158, +0.185]), up from the transductive +0.110. The off-lineage AUROC collapses toward chance on this basis while the in-lineage diagonal stays high, and a label-shuffle permutation null is -0.001, so the diagonal bonus is not a fitting artifact. As a further pipeline negative control, splitting each family into two pseudo-families by a random partition of its task IDs and running the identical strict head-fitting pipeline gives a diagonal interaction of +0.000 (95% band [-0.012, +0.012]) across 300 partitions per family, with no partition reaching the observed +0.172: the pipeline manufactures no diagonal bonus when there is no true family boundary. The eight-family and nine-family interactions are +0.197 ([+0.188, +0.206]) and +0.194 ([+0.185, +0.202]) leak-free (Table 2b). We report the leak-free strict basis as

Roster / cut	Transductive (basis leaks target family)	Leak-free (strict, primary)
APPS, 4 open families, full	+0.110 [+0.094, +0.125]	+0.172 [+0.158, +0.185]
APPS, 4 open families, action-only	+0.055 [+0.045, +0.065]	+0.079 [+0.065, +0.092]
APPS, 5 families (+1 frontier vendor), full	+0.132 [+0.122, +0.143]	+0.194 [+0.182, +0.206]
APPS, 5 families (+1 frontier vendor), action-only	+0.069 [+0.058, +0.079]	+0.095 [+0.081, +0.109]
APPS, 6 families (2 frontier vendors), full	+0.137 [+0.128, +0.146]	+0.195 [+0.184, +0.205]
APPS, 6 families (2 frontier vendors), action-only	+0.120 [+0.110, +0.132]	+0.122 [+0.109, +0.136]
APPS, 8 families (4 frontier vendors), full	+0.150 [+0.142, +0.157]	+0.197 [+0.188, +0.206]
APPS, 9 families (5 frontier vendors), full	+0.152 [+0.145, +0.159]	+0.194 [+0.185, +0.202]
APPS, frontier-only (4 vendors), full	+0.103 [+0.094, +0.110]	+0.10 (unchanged)
bigcodebench, 3 families, full	+0.250 [+0.233, +0.268]	+0.233 [+0.214, +0.251]
bigcodebench, 3 families, action-only	+0.141 [+0.127, +0.155]	+0.099 [+0.085, +0.114]

Table 2b: Leak-free strict re-estimation (scaler, PCA, and head fit on the calibration family alone, evaluated family absent from the pipeline). The strict basis removes target-family leakage from the shared preprocessing and is larger than or unchanged from the transductive estimate on the main APPS rosters shown, because the pooled basis was diluting the diagonal advantage there. The BigCodeBench substrate check is lower on the strict basis, and is reported as the one exception. The only roster still reported on the transductive basis is SHADE (non-code, reported as underpowered); the LoRA monitor is family-conditioned by construction and reported in Table 2. The strict full-transcript interaction is monotone in family count and then plateaus: +0.172 (four open), +0.194 (five), +0.195 (six), +0.197 (eight), +0.194 (nine).

primary; the transductive numbers elsewhere in this section and in the APPS rows of Table 2 are a conservative lower bound.

Successful-attacker robustness. We first restrict the evaluation to attacks that completed the main task. On that competent-attacker slice the leak-free interaction is +0.169 ([+0.142, +0.196]) on the four open families and +0.201 ([+0.185, +0.216]) on the eight-family matrix, indistinguishable from the unfiltered +0.172 and +0.197. Restricting to attacks that did not simply fail neither inflates nor deflates the ranking advantage, so the advantage is about how a lineage’s attack is written, not about which attacks happened to fail.

We can go one step further on the lineages that realize sabotage. Realized sabotage (the main task passing and the backdoor firing on the secret input) is frontier-only on this substrate: the four open families produce essentially no clean backdoors, so a full realized-sabotage matrix is not powerable across them (Section 4.4). Restricting to the four frontier families that do realize it (GPT-5 powered to 50 test both-success positives; Grok to 41 after exhausting its entire APPS task pool; DeepSeek and Sonnet thinner at 15 and 7), the calibration-family interaction survives on realized backdoors. Because both the diagonal and the off-diagonal AUROC saturate near 1.0 here, every head detects a successful backdoor well, AUROC compresses to +0.030 ([+0.018, +0.044], tightened from [+0.013, +0.060] when Grok had 11 positives); the non-saturating effect size (Cohen’s d on the head scores) is +0.738 ([+0.407, +1.027]), about 1.1 times the attempt-level effect size on the same families. The family-specific signature is therefore at least as strong in backdoors that actually fired, while leaving the operational catch-rate gap modest on realized sabotage, since foreign

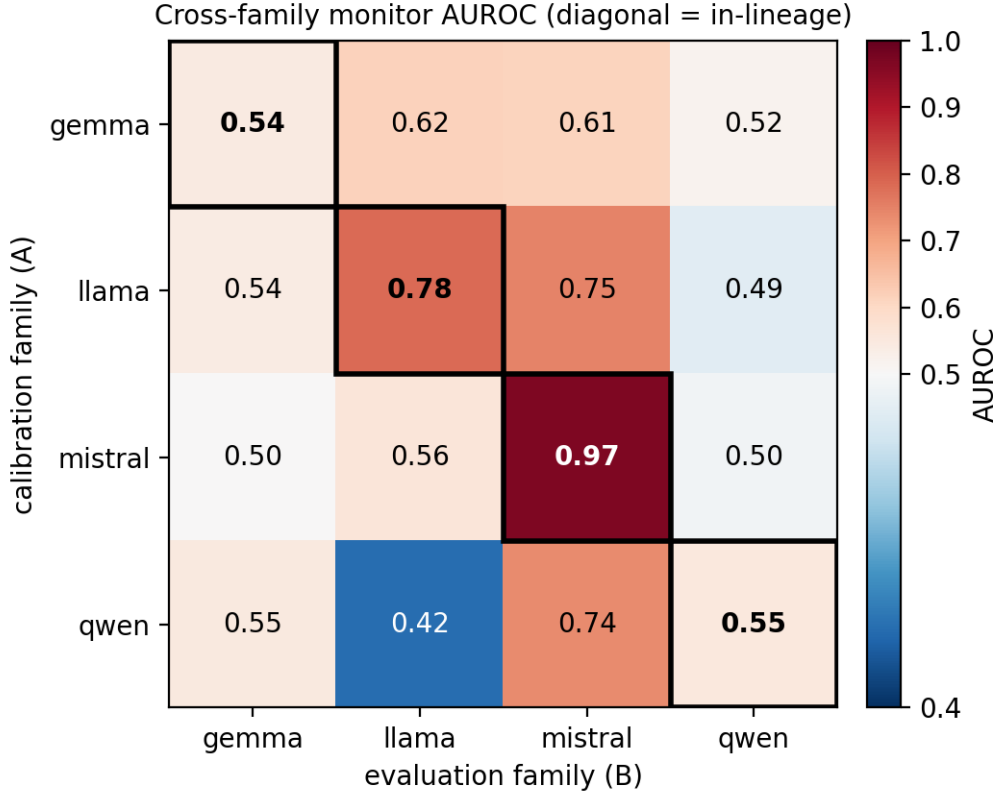


Figure 1: Cross-family monitor AUROC, four open-weight families (diagonal = in-lineage).

monitors still detect successful backdoors well. We report this as a robustness slice rather than a powered headline because only GPT-5 reaches the preregistered 50-positive gate.

The contrast that makes this a measurement claim rather than an artifact is the fixed-prompt baseline. A fixed-prompt monitor has no calibration-family ranking axis, so its only calibration-dependence is where the 1%-FPR threshold lands. Its Track-1 diagonal bonus is null at every threshold we tested (0.5%, 1%, 5%). The shift Track-2 measures is a ranking property of the family-conditioned monitor, not a threshold-placement artifact.

4.2 Monitor-class robustness

A natural objection is that the interaction is specific to a frozen-embedding logistic head. It is not. The effect appears in a trained transcript-reading monitor, survives an order of magnitude of monitor scale, and does not depend on the choice of head.

The decisive evidence is a trained monitor. We finetuned a LoRA adapter on microsoft/Phi-3.5-mini, a base vendor-disjoint from all four policy families, per calibration family, reading the raw transcript rather than a frozen embedding. It cleared the competence bar (Mistral diagonal 0.999, Llama diagonal 0.757, close to the embedding head’s 0.74) and replicated the interaction more strongly, at +0.179 (95% CI [+0.156, +0.201]). The number is robust to training randomness and base choice: two further seeds give +0.204 and +0.194, and a second vendor-disjoint base (the 3.8B Phi-4-mini, a distinct model from the 14B Phi-4 below) gives +0.180, every CI excluding zero and every run clearing the bar.

That monitor is small, and the sharpest form of the objection is that a real deployer finetunes a strong, frontier-capability monitor that would see past family-specific surface style, making the gap an artifact of monitor weakness. We tested it with a capability staircase that holds the training procedure, the split, the labelled data, the epochs, and the LoRA

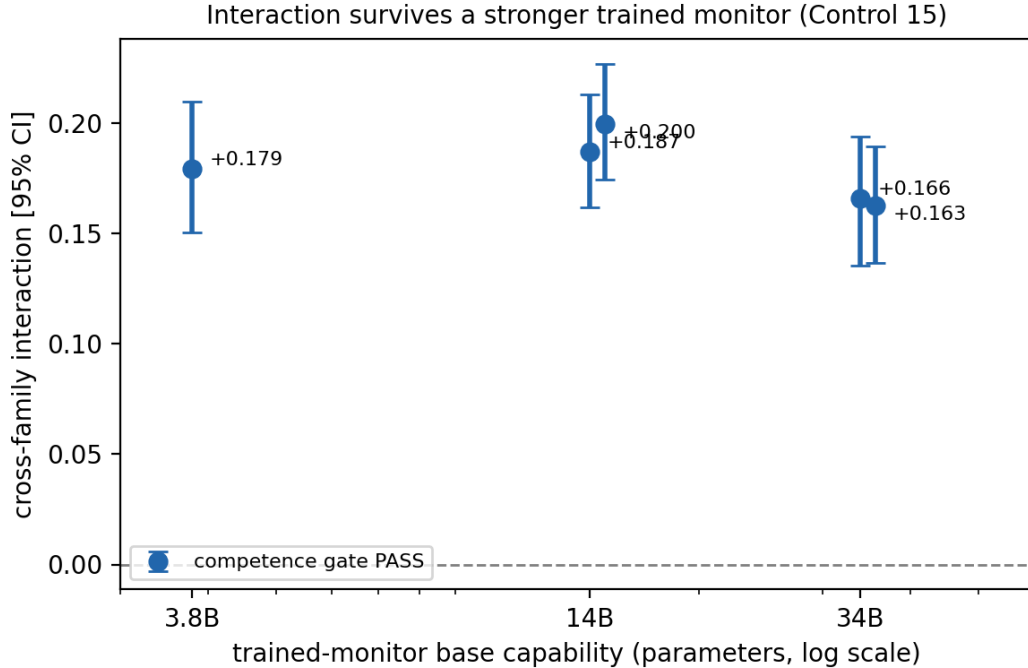


Figure 2: Monitor-capability staircase: the cross-family interaction stays positive (CI excluding zero) at every trained-monitor scale that clears the competence gate, from Phi-3.5-mini (3.8B) through Phi-4 (14B) to Yi-1.5 (34B). Scale does not close the gap.

rank fixed and varies only base-model scale, from Phi-3.5-mini (3.8B) through Phi-4 (14B) to Yi-1.5-Chat (34B), a roughly nine-fold range. The interaction stays positive with a CI excluding zero at every rung that clears the competence gate: +0.179 at 3.8B, +0.187 and +0.200 at 14B across two seeds, and +0.166 [+0.143, +0.187] and +0.163 [+0.143, +0.182] at 34B, the largest monitor itself clearing the gate (Mistral diagonal 1.000, Llama 0.776). An order of magnitude of added capability neither closes the gap nor trends it toward zero; the overfit is structural (Figure 2).

The head does not matter either. Swapping the logistic head for an MLP, an RBF SVM, a random forest, or gradient boosting leaves the interaction positive in every case (0.111 to 0.159), so the overfit lives in the family-conditioned representation rather than the linear head.

The one monitor class we could not train is the prompted monitor practitioners actually deploy, and we probed it three ways without any clearing the competence bar. Four in-context few-shot monitors, spanning mid-tier, reasoning, and frontier capability across two vendors (gpt-4.1-mini, o4-mini, gpt-4.1, and the frontier Grok-4.3), all rated Llama’s attacks below its honest code; sixteen exemplars convey too little family signal, and a monitor that cannot rank a family’s sabotage at all cannot overfit its ranking. To rule out that the exemplar wiring rather than the prompted regime was the problem, we then built the strongest static prompted scaffold short of training: a detection rubric distilled from each calibration family’s own attacks plus chain-of-thought on gpt-4.1, which gives the prompted monitor a genuine calibration-family axis. It too fails the bar: only Mistral’s diagonal clears 0.70 (0.751), while Llama (0.419), Qwen (0.531), and Gemma (0.590) sit at or near chance (its overall interaction is positive, +0.040, in the same direction as the trained monitors, and the competent Mistral row shows the same diagonal bonus, but a monitor that ranks three of four families at chance cannot adjudicate the question). Third, we built the scaffold a deployer holding a pool of past labeled transcripts would actually run: per-query retrieval, placing the k=8 nearest benign and k=8 nearest backdoored calibration-family transcripts (text-embedding-3-small cosine) in context and instructing the judge to rate by relative

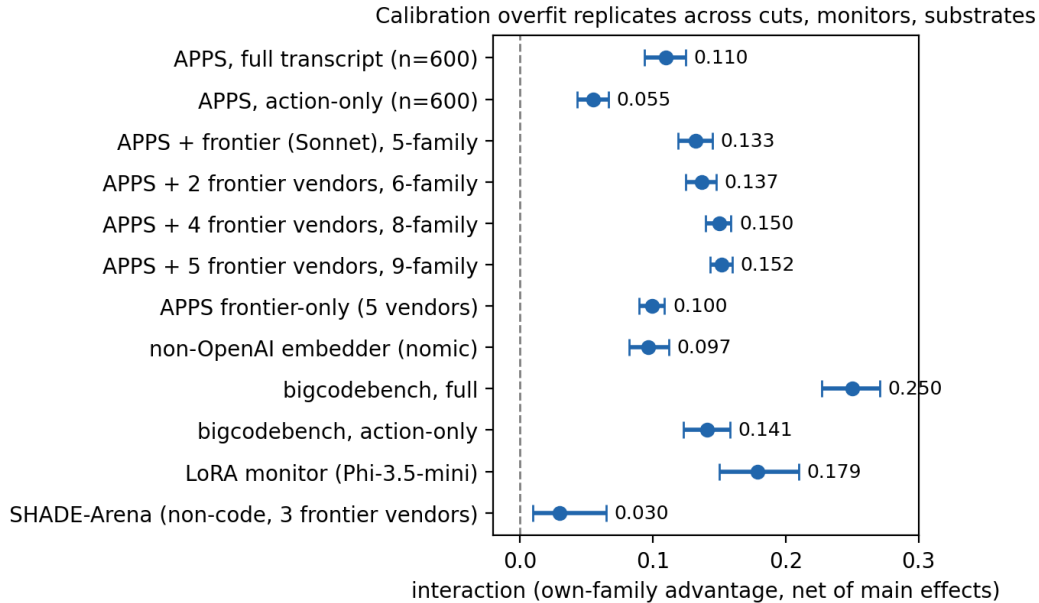


Figure 3: The interaction replicates across cuts, monitor classes, and substrates (95% CIs).

resemblance to them rather than an absolute rubric. It still fails on Llama: gpt-4.1-mini diagonal 0.300 and frontier gpt-4.1 diagonal 0.381, both below chance, against Mistral 0.745 and 0.843 which pass. This probe is decisive because it separates the signal from the judge: a retrieval-only kNN-margin scorer over the same vectors handed to the judge ranks Llama at 0.729 and Mistral at 0.953 and reproduces the cross-family interaction at +0.130 (95% CI [+0.108, +0.153], cluster-bootstrap by task), matching the trained embedding head’s +0.127 and preserving the same Mistral-to-Llama anti-transfer (0.396, below chance). The family-conditioning information is therefore provably present in context; the prompted judge cannot convert it into a Llama ranking, reverting to the absolute red-flag judgment that rates Llama’s clean backdoors below its honest code. Across six prompted constructions (four few-shot, one rubric-distilled, one retrieval-augmented over two judges) every prompt-only family-conditioned monitor fails Llama at or below chance. We therefore read the prompted class as a clean negative rather than merely inconclusive: in-context family conditioning, even with per-query retrieval and a frontier judge, is insufficient, and the interaction is demonstrable only with a trained monitor. We accordingly scope the established claim to trained, family-conditioned monitors (Section 7). One caveat: the two LoRA bases share a vendor (Microsoft), though both are vendor-disjoint from all four policy families (Section 7).

The trained-monitor result extends to the closed-API fine-tuned monitor a deployer would actually procure. Per-family supervised fine-tunes of gemini-2.5-flash (Vertex AI, identical balanced calibration splits, single-token backdoor label, three epochs) clear the preregistered competence gate on both gated families, with Mistral diagonal 0.738 and Llama 0.706, and the cross-family interaction survives at +0.100 (95% CI [+0.063, +0.134], cluster-bootstrap by task; mean diagonal 0.627 against off-diagonal 0.495), consistent with the open trained monitors. The tuned Llama monitor ranks Mistral’s sabotage below chance (0.405), reproducing the anti-transfer signature of the embedding head. Qwen and Gemma diagonals (0.535, 0.531) sit under the gate and are read descriptively; closed tuned weights are not releasable, so the open ladder remains the reproducible backbone (Section 7).

4.3 External validity: other substrates

We regenerated three families (Mistral, Llama, Qwen) on bigcodebench, a different code distribution, using the identical pipeline (same split, scaler, PCA, head, bootstrap), and re-ran

the matrix. The interaction replicates leak-free: +0.233 (95% CI [+0.214, +0.251]) on the full transcript and +0.099 ([+0.085, +0.114]) code-only on the strict basis (the transductive basis reads +0.250 / +0.141; unlike APPS the leak correction lowers the bigcodebench estimate, so the strict number is the conservative one). The anti-transfer signature is clear: foreign heads score Llama’s bigcodebench attacks below chance (Mistral-head 0.18, Qwen-head 0.17), while Llama’s own head scores them at 0.86. An independent 2-family replication on separately generated transcripts gave +0.244 [+0.234, +0.254], corroborating the 3-family number.

The bigcodebench generation also surfaced an infrastructure point worth recording: the Llama family completed cleanly on a native amd64 host but had failed under x86 emulation, which is a performance artifact rather than a model limitation.

SHADE-Arena (Kutasov et al., 2025b) gives a non-code test of the same question: agentic tool-use sabotage in mock workplace environments, a transcript distribution with nothing in common with code backdooring. We use the gated task set under a data agreement; it is held out of pre-training corpora, so we report aggregate numbers only, withhold all task detail, and honor the dataset’s canary string. Trajectories are the dataset’s bundled originals from three frontier vendor lineages (Claude 3.5 Sonnet, GPT-4.1, Gemini 2.5 Flash) on 16 task pairs, deduplicated, with the system prompt and hidden scratchpad stripped from the monitor-visible text, the same embedding monitor and split machinery as everywhere else in this paper, and a pre-registered primary cut (full transcript, all attempts). The interaction is directionally consistent at +0.035 (cluster-bootstrap 95% CI [-0.004, +0.077]), about a third of the code-substrate magnitude but with an interval that includes zero, on a much harder absolute problem for this monitor class (mean diagonal AUROC 0.53 versus roughly 0.75 on code). The interval spans zero from power, not absence: the primary cut has only seven test tasks, giving a minimum detectable effect of +0.054 at 80% power, above the observed point estimate, so the public split cannot resolve an effect of this size. The mechanism evidence lines up: family identity is only weakly recoverable from SHADE transcripts (1.2x chance, versus 2.6x on APPS), and on the action trace it is unrecoverable (chance accuracy), where the interaction is also null (-0.021 [-0.067, +0.032]). SHADE’s action-only canonicalization is the agent’s tool-call trace, a different construction from the submitted-code action-only cut on APPS, so the two action-only numbers are not directly comparable and we do not equate them. Success-filtered cuts are null with intervals wider than the primary effect, so they neither confirm nor contradict it. A with-hint attack variant, where agents receive a side-task strategy hint and coverage is heavily family-imbalanced, is null (-0.006 [-0.031, +0.027]); the null is not explained by hints homogenizing execution (family identity is more recoverable from hinted attacks, 1.4x versus 1.1x chance), so we scope the replication to the pre-registered unhinted mode and flag the hint mode as an open boundary. Fixed monitors show the predicted null diagonal bonus (three monitors, every CI spanning zero), and the static-threshold failure replicates: realized off-lineage audit spend runs 0 to 8.6% against a 1% target for the competent monitors. At a 0.53 diagonal AUROC the budget-enforced protocol resolves no realized-safety gap on this substrate. We read SHADE as directionally consistent but underpowered: the point estimate carries the code substrate’s sign at about a third its magnitude, but at seven test tasks we report it as suggestive non-code evidence rather than a significant replication. The behavioral residual and the realized-safety cost we measure on code do not appear on this substrate at this sample size; a full SHADE split, with roughly three times the task clusters, would be needed to resolve whether the corrected interval clears zero.

4.4 Realized safety: what the protocol delivers

AUROC is the right metric for the mechanism but not the deployment claim. We translate the interaction into the number a defer-to-trusted protocol delivers under an audit budget q , the fraction of benign traffic the blue team can afford to flag. Control deployments use q near 1%.

The deployment-correct protocol audits the top- $q\%$ most-suspicious actions in the live stream. Because attacks are rare, this holds the realized false-positive rate at q on every family by construction, which isolates the ranking failure: at matched audit spend, a

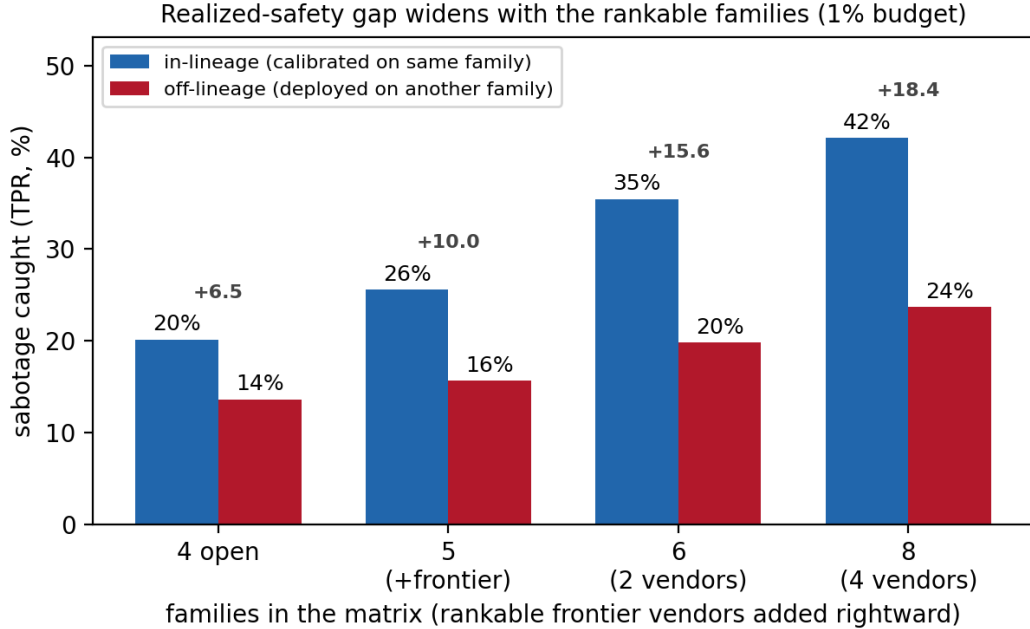


Figure 4: Realized catch rate under a defer-to-trusted protocol at a 1% audit budget, as rankable frontier families are added to the matrix (four open-weight to eight families). At eight families the headline strict leak-free pair is 41% in-lineage versus 19% off-lineage, a +21.7-point gap; the curves shown use the transductive basis, on which the in-lineage advantage widens from +6.5 to +18.4 points (42% versus 24%), and the off-lineage curve rises far more slowly.

Table 3: Realized catch rate of a defer-to-trusted protocol at fixed audit budgets, in-lineage versus off-lineage (budget-enforced, false-positive rate held at q on every cell). These rates use the transductive basis; the leak-free strict re-estimation is in Table 3b and widens the gap, because the strict off-lineage ranking is nearer chance.

Audit budget q	Families	In-lineage	Off-lineage	Gap (95% CI)
1%	4 open	20%	14%	+6.5 [+4.5, +7.8]
1%	5 (+ frontier)	26%	16%	+10.0 [+7.2, +13.7]
1%	6 (2 frontier vendors)	36%	20%	+15.6 [+12.4, +18.5]
2%	4 open	21%	15%	+6.2
2%	5 (+ frontier)	30%	18%	+12.2
1%	8 (4 frontier vendors)	42%	24%	+18.4 [+15.8, +21.1]
5%	4 open	26%	18%	+8.8
5%	5 (+ frontier)	35%	21%	+13.7
5%	8 (4 frontier vendors)	52%	31%	+20.2

difference in catch rate is a difference in ranking. We verified the false-positive rate sits at q on every cell (1.3% at $q=1\%$, range 1.2 to 1.4%).

Table 3 reports the catch rate under this protocol. At a 1% budget on the four open families the in-lineage monitor catches 20% of attack attempts and an off-lineage monitor 14%, a 6.5-point gap (an own-versus-foreign ratio of about 1.5x); adding one frontier family widens this to 26% versus 16%, a 10.0-point gap, and the full eight-family matrix with four frontier vendors widens it to 42% versus 24%, an 18.4-point gap (a 1.8x ratio). The gap is the main-effects-removed interaction on catch rates and is stable across budgets.

Audit budget q	Families	Leak-free gap (95% CI)
1%	4 open	+16.9 [+14.0, +21.8]
2%	4 open	+20.1 [+16.2, +24.2]
5%	4 open	+23.8 [+20.1, +27.0]
1%	8 (4 frontier vendors)	+21.7 [+19.7, +26.7]
2%	8 (4 frontier vendors)	+26.8 [+22.4, +28.9]
5%	8 (4 frontier vendors)	+29.0 [+26.3, +31.3]
1%	9 (5 frontier vendors)	+24.8 [+22.5, +29.1]
2%	9 (5 frontier vendors)	+29.2 [+25.0, +31.0]
5%	9 (5 frontier vendors)	+30.2 [+27.8, +32.3]

Table 3b: Leak-free strict realized-safety gap (diagonal minus off-diagonal catch rate, net of main effects) at three audit budgets, budget-enforced with the false-positive rate held at q on every cell.

Leak-free re-estimation. On the strict leak-free basis the off-lineage catch rate falls further, so the safety gap is larger than the transductive table reports. At a 1% budget on the four open families the in-lineage monitor catches 18% of attack attempts while the off-lineage monitor catches under 2%, a +16.9-point gap ([+14.0, +21.8]); on the eight-family matrix the split is 41% versus 19%, a +21.7-point gap ([+19.7, +26.7]); on nine families it is 47% versus 22%, a +24.8-point gap ([+22.5, +29.1]). The false-positive rate sits at 1.3% on every cell (1% target), and the gap is monotone in the budget. These supersede the transductive numbers for the headline; the qualitative claim is unchanged and stronger, an off-lineage attacker is roughly twice as likely (and on the open families far more likely) to slip past auditing at the same budget as the in-lineage attacker the monitor’s published number was measured on.

Per-family view: which lineages are actually caught. The mean gap is not uniform; it is carried by the families a monitor can rank at all. At a 1% budget on the four open families (transductive basis; the strict basis widens the spread further):

Evaluation family	Own-lineage catch	Foreign-lineage catch	Own - foreign
Mistral	77.6%	30.7%	+46.9
Llama	1.3%	2.6%	-1.3 (inverts)
Qwen	3.7%	4.1%	-0.4
Gemma	1.2%	1.2%	+0.0

Mistral’s own head catches 78% of its sabotage at a 1% budget where foreign heads catch 31%; for Gemma and Qwen both rates sit at the budget floor, where the monitor barely ranks the family at all; and for Llama the own head is no better than foreign. The mean interaction is the honest single-number summary because it is net of the Mistral row and column main effects and the task-clustered bootstrap accounts for the concentration, but the per-family reading is precise: for the lineages a family-conditioned monitor can rank at deployment budgets, calibrating off-lineage roughly halves attack-attempt catch at identical audit cost.

A secondary finding strengthens the deployment story. If, instead of re-ranking the live stream, the blue team ships a static threshold calibrated on family A, the protocol does not even hold its audit budget off-lineage: realized audit spend on family B ranges from 0% to 94% (mean 27%) against a 1% target, because each family’s benign score distribution sits in a different place. The operating point itself is family-specific. The two address different questions: the budget-enforced protocol measures ranking transfer at matched audit cost, while the static threshold measures operating-point transfer. We report the budget-enforced number as the headline because it is the apples-to-apples comparison at fixed cost, and the static-threshold failure as the additional warning that the operating point does not transfer either.

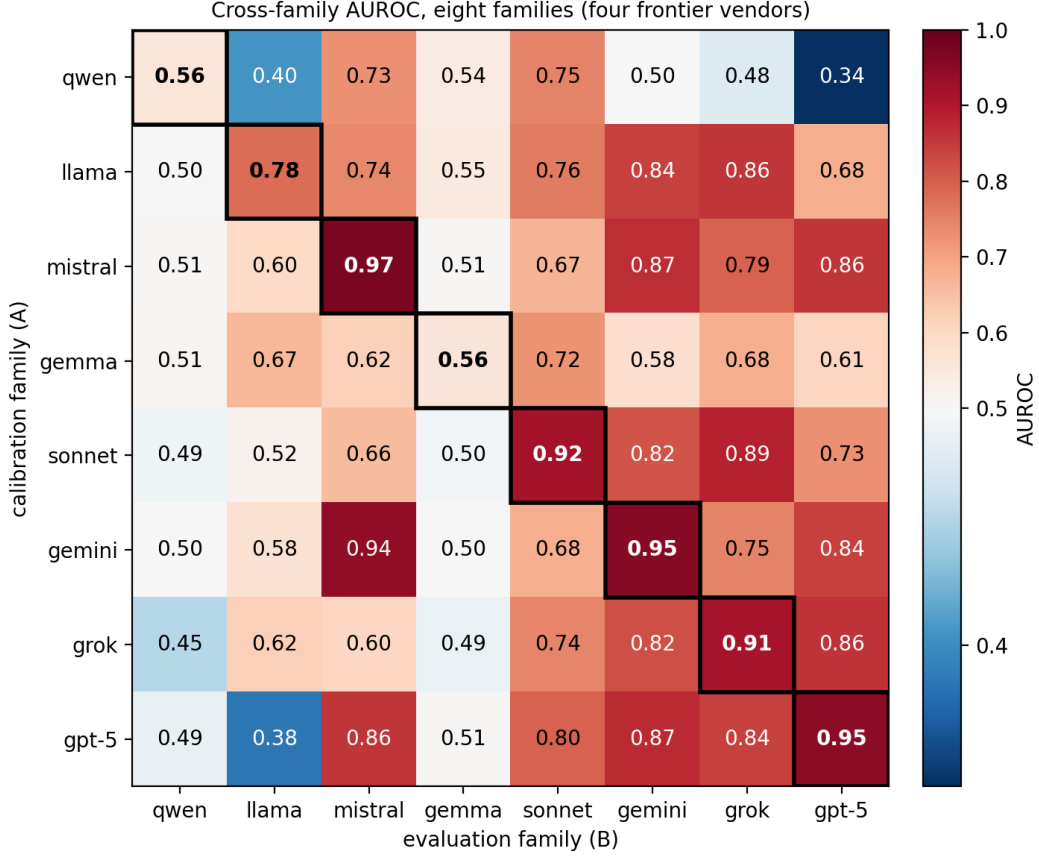


Figure 5: Cross-family monitor AUROC across all eight families (four open-weight plus four frontier vendors: Anthropic Sonnet, Google Gemini, xAI Grok, OpenAI GPT-5). Rows are the calibration family, columns the evaluation family; the bold-bordered diagonal is in-lineage. The own-family advantage net of attacker-obviousness (column) and monitor-capability (row) main effects, the interaction, is $+0.150$.

The catch-rate gap at a 1% budget is carried by the lineages a monitor can rank at all (Section 7): at this budget the in-lineage catch rate is high for Mistral and modest for the weaker attackers, so the mean gap reflects rankable lineages rather than all families. The behavioral (code-only) cut shows no catch-rate gap at deployment budgets, consistent with its smaller AUROC interaction ($+0.055$) being below the resolution of a 1% budget even at 600 transcripts per family.

These catch rates are for detecting attack attempts, not realized backdoors. Both-success backdoors (the main task passes and the backdoor fires on the secret input) occur in under about 5% of attacks and are near zero for the sub-frontier open families, so an all-family realized-sabotage matrix is not computable at this scale: in the eight-family safety matrix, 5 of 8 families have 0 to 1 successful-backdoor test positives, below any usable per-cell count. In the later frontier realized-sabotage slice (GPT-5, DeepSeek, Grok, Sonnet) the interaction survives as an effect size at least as large as at the attempt level (Section 3), but only one of those cells (GPT-5) reaches the preregistered 50-positive gate. We therefore report the gap for attack-attempt detection as the primary axis and the realized-sabotage cut as a robustness slice, and flag the inability to power a full realized-backdoor matrix at this substrate as a limitation (Section 7); closing it would require generating a substrate with a higher clean-backdoor yield.

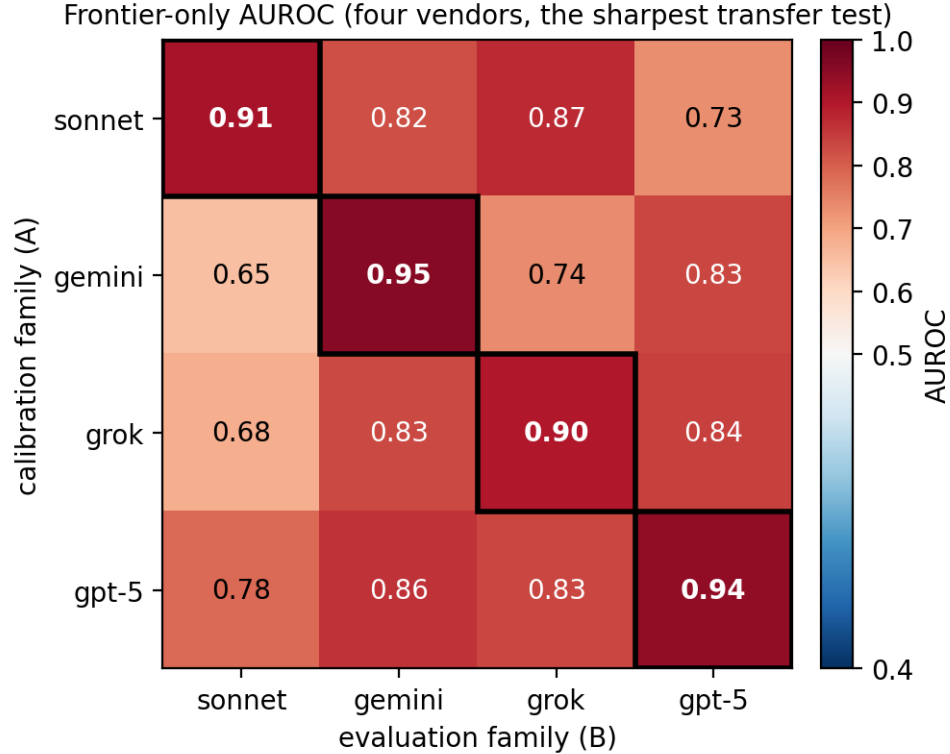


Figure 6: Frontier-only sub-matrix: four strong models from four independent vendors, the sharpest transfer test, where a capable monitor should transfer best. Every model’s own head ranks its sabotage best (bold-bordered diagonal, 0.90 to 0.95) while foreign heads fall to 0.65 to 0.87; the interaction is +0.103 (full transcript) and +0.135 (action-only).

4.5 Frontier attackers across five vendors

The four headline families are open-weight and sub-frontier. To test whether the interaction is an artifact of attacker weakness, we generated a fifth family with a frontier closed-API model, Claude Sonnet 4.6, on the identical APPS setting ($n=600$, native provider, vendor-disjoint from the gpt-4.1-mini monitor and from all four open families). It attacks competently: zero refusals across 600 attack samples, 68% main-task success, and 13 clean both-success backdoors (main passes and the secret input fails) where the open families produce almost none.

Adding the frontier family strengthens, rather than dilutes, the interaction: +0.132 (95% CI [+0.122, +0.143]) on the full transcript and +0.069 ([+0.058, +0.079]) action-only (Table 2).

The frontier family shows the diagonal-overfit signature in both directions: its own head ranks its attacks at 0.92 AUROC, while the Sonnet-calibrated head collapses to at or below chance on the open families (Llama 0.48, Qwen 0.48, Gemma 0.51). The realized-safety gap also grows, to +10.0 points at a 1% budget (95% CI [+7.2, +13.7]), because the frontier family is a second strongly-rankable lineage with a large own-versus-foreign catch-rate difference, so the gap is no longer carried by Mistral alone. One qualification: the Sonnet column is moderately detectable by foreign heads (0.65 to 0.78), so the frontier model’s backdoors carry a real attacker-obviousness component; the two-way decomposition removes that column effect and the diagonal advantage still strengthens. The pre-registered dilution risk, a ceiling or floor column that washes out the diagonal, did not occur.

Table 4: Five-family cross-family AUROC (rows = calibration family, columns = evaluation family; in-lineage diagonal in bold). The diagonal dominates the row for the three rankable lineages (Sonnet, Mistral, Llama); the two weakest attackers (Gemma, Qwen) have near-chance own-family diagonals (0.55, 0.56) that sit below their cells on the obvious Sonnet column, the Section 7 “rankable lineages” limitation made visible. The interaction removes that column effect, so the diagonal advantage it estimates, net of that column effect, is positive even for the weak rows. Foreign heads fall to or below chance on several lineages (Qwen-on-Llama 0.43, Sonnet-on-Llama/Qwen 0.48).

Calibrated on / Evaluated on	Sonnet	Mistral	Llama	Gemma	Qwen
Sonnet	0.92	0.62	0.48	0.51	0.48
Mistral	0.65	0.97	0.56	0.51	0.50
Llama	0.76	0.73	0.78	0.54	0.49
Gemma	0.78	0.60	0.63	0.55	0.51
Qwen	0.72	0.71	0.43	0.55	0.56

A second frontier vendor. To check that the frontier result is not specific to one vendor, we added Google Gemini 2.5 Flash as a sixth family, vendor-disjoint from the monitor and from Sonnet. With both frontier vendors in, the interaction is +0.137 (95% CI [+0.128, +0.146]) on the full transcript and +0.120 ([+0.110, +0.132]) action-only, and the realized-safety gap grows to +15.6 points at a 1% budget ([+12.4, +18.5]). Gemini’s own head ranks its attacks at 0.95 AUROC while foreign heads collapse on them once transcript length is stripped (Llama 0.30, Mistral 0.38). One methods caveat: Gemini does not call submit early in the agentic loop, so for tractability it ran at a reduced step cap (5 versus 15), giving shorter transcripts; we therefore treat the action-only interaction (+0.120, length-stripped, which is in fact larger than the five-family action-only +0.069) as the confound-robust read for the six-family matrix. The own-family advantage thus holds across two independent frontier vendors.

Two more frontier vendors: xAI and OpenAI. We then added two further frontier lineages from independent vendors, xAI Grok and OpenAI GPT-5, bringing the matrix to eight families that span four frontier vendors (Anthropic, Google, xAI, OpenAI) alongside the four open-weight families. The interaction rises to +0.150 (95% CI [+0.142, +0.157]) on the full transcript and +0.138 ([+0.126, +0.150]) action-only. Both new attackers are competent and each ranks its own sabotage near-perfectly (Grok 0.91, GPT-5 0.95 own-head AUROC), a large attacker-obviousness column that the two-way decomposition removes, leaving the diagonal advantage larger rather than smaller. An exact permutation test that holds the row and column main effects fixed and enumerates all 40,320 family matchings ranks the observed diagonal first in every case ($p = 0.000025$, full and action-only): the own-family matching is the single most extreme of all matchings, not merely a stable point estimate. The sharpest version of the calibration-overfit test is the frontier-only sub-matrix of these four vendors alone, where every attacker is a strong model and a capable monitor should transfer best. Even there the interaction is +0.103 (95% CI [+0.094, +0.110]) on the full transcript and +0.135 ([+0.120, +0.150]) action-only; the code-only cut being the larger of the two rules out a transcript-length or weak-open-model artifact and is the behavioral form of the claim. Translated into a defer-to-trusted protocol at a 1% budget, the eight-family realized-safety gap is +21.7 points on the strict leak-free basis (41% in-lineage versus 19% off-lineage; on the transductive basis +18.4 points, 42% versus 24%, [+15.8, +21.1]). Two caveats: GPT-5 shares its vendor with the gpt-4.1-mini fixed-prompt monitor, but every number here comes from the embedding head, whose vendor-independence we verify by reproducing the interaction with a non-OpenAI open-weight embedder on the four-open matrix (Section 6); the embedder is itself OpenAI, but a same-vendor effect could only help a same-vendor monitor detect and so cannot inflate a transfer gap; and GPT-5’s attack transcripts completed at a lower yield (325 successful attacks versus roughly 600 for the other

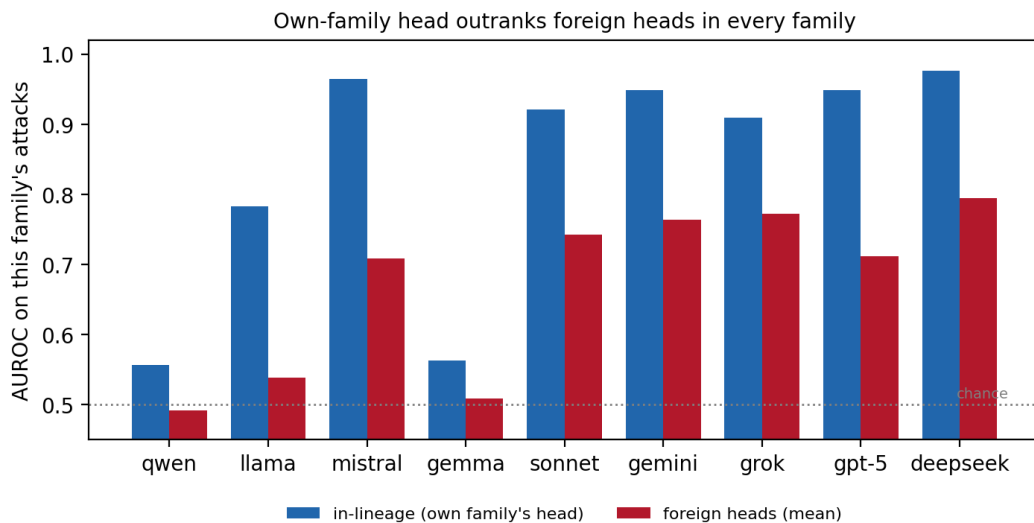


Figure 7: In-lineage (own-family head) versus mean foreign-head AUROC per family, all nine families: the own-family head outranks the foreign heads for every family, sharply for the rankable lineages and by a near-chance margin for the weakest attackers.

families), so its individual cells carry wider intervals, which the task-clustered bootstrap already reflects.

A fifth frontier vendor: DeepSeek. Adding DeepSeek-V3.2 (thinking disabled), a fifth independent frontier lineage and one vendor-disjoint from the gpt-4.1-mini monitor, brings the matrix to nine families. The interaction is $+0.152$ (95% CI $[+0.145, +0.159]$) on the full transcript and $+0.157$ ($[+0.146, +0.167]$) action-only. This is the largest action-only interaction we report and is now larger than its own full-transcript value: the behavioral component of the effect strengthens, not weakens, as more frontier lineages enter. DeepSeek ranks its own sabotage almost perfectly (0.978 own-head AUROC, the most detectable attacker in the set), another large attacker-obviousness column that the decomposition removes. The exact permutation test over all 362,880 nine-family matchings again ranks the observed diagonal first ($p = 0.000003$, full and action-only). The frontier-only sub-matrix, now five vendors, holds the interaction at $+0.100$ ($[+0.092, +0.107]$) on the full transcript and $+0.164$ ($[+0.150, +0.177]$) action-only; with five families the permutation test is resolvable below 0.05 on its own (rank 1 of 120, $p = 0.0083$, where the four-vendor floor was 0.042), so the frontier-only effect is now significant by permutation as well as by bootstrap, and the action-only cut remains the larger of the two. Translated into a defer-to-trusted protocol at a 1% budget, the nine-family realized-safety gap on the transductive basis is +20 points (47% in-lineage versus 26% off-lineage, $[+17.9, +23.1]$); the leak-free re-estimation of Section 4.4 widens it to +24.8 as the off-lineage rate falls to 22%.

4.6 Calibrating on weaker models, deploying against a frontier attacker

The most deployment-relevant version of the transfer question is a capability gap: a blue team calibrates its monitor on the models available today and deploys it against a stronger model it did not have. The five families in this reanalysis span a clear capability range (attack-mode main-task success: Qwen 0.32, Gemma 0.39, Mistral 0.40, Llama 0.41, and the frontier Claude Sonnet 0.68), so the cells that calibrate on an open family and evaluate on Sonnet are exactly this scenario.

Monitoring the frontier from weaker-model calibration roughly halves detection. A monitor calibrated on Sonnet ranks Sonnet’s own attacks at 0.92 AUROC; the open-family heads average 0.73 on the same attacks, a gap of $+0.189$ (95% CI $[+0.172, +0.205]$). Under the

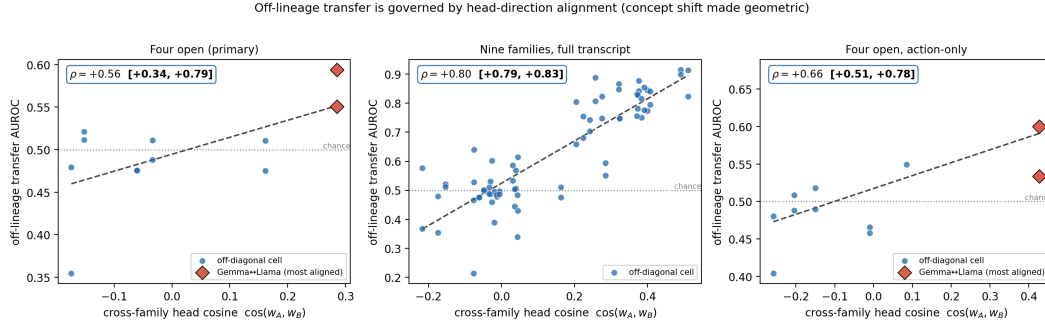


Figure 8: Off-lineage transfer AUROC rises with the cosine alignment between the calibration and evaluation families’ sabotage directions, one point per off-diagonal cell. Left: four open families. Middle: nine families. Right: four open, action-only.

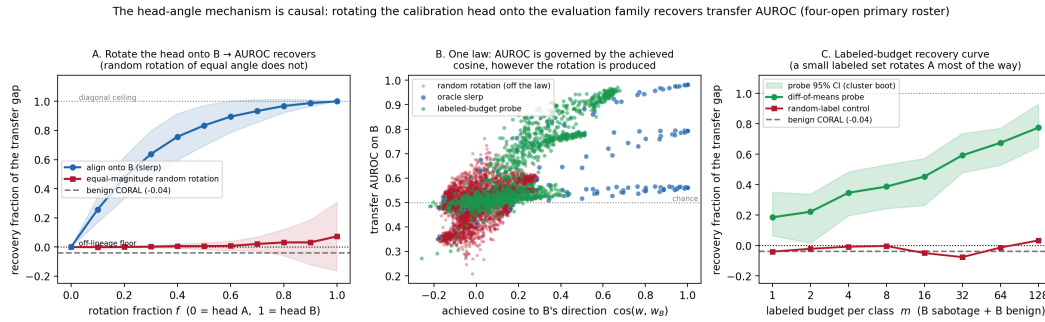


Figure 9: The head angle is a causal lever (four-open primary roster). Left: rotating calibration head A onto evaluation family B recovers transfer AUROC monotonically, while an equal-magnitude random rotation stays flat and benign-only CORAL alignment falls below the off-lineage floor. Middle: oracle-rotation and labeled-budget-probe points obey one cosine-to-AUROC law; random rotations sit off it. Right: a difference-of-means direction probe recovers the gap as the labeled budget from B grows, while a label-shuffled probe of equal size recovers nothing.

1% audit budget the in-lineage monitor catches 47% of Sonnet’s sabotage and the weaker-calibrated monitors catch 24%.

A capability ordering does not explain the failure. A naive weak-to-strong reading predicts that calibrating on weaker models and testing on a stronger one is the hard direction. We find the opposite: weak-to-strong transfer averages 0.64 AUROC while strong-to-weak averages 0.54, an asymmetry of 0.10 (95% CI [0.09, 0.11]) in favour of weak-to-strong, and capability distance does not predict worse transfer (the correlation between absolute capability gap and off-diagonal AUROC is +0.21, the wrong sign for a capability-distance story). The Sonnet-calibrated head is the most family-overfit of all, ranking the open families at or below chance, while the frontier’s attacks retain enough cross-visible obviousness that weaker-calibrated heads still partly catch them. So the deployment cost is real and large, but it is driven by which family the monitor was calibrated on, not by the monitor being too weak for a stronger attacker. This quantifies the capability-gap deployment risk and distinguishes the result from weak-to-strong monitoring work, whose axis is monitor capability rather than calibration family.

4.7 Mechanism: misaligned sabotage directions

The gap is geometric. To characterize the concept shift behind the off-lineage collapse, we read each family’s sabotage-discriminative direction directly from its strict head, the gradient of its decision function in embedding space, w_A . If the transfer gap were a covariate shift

these directions would agree and only the operating point would move; instead they disagree, and the disagreement predicts the damage. Across off-diagonal cells, the transfer AUROC of head A on family B rises with the cosine similarity $\cos(\mathbf{w}_A, \mathbf{w}_B)$: Spearman rho is +0.56 on the four-open primary matrix (95% CI [+0.34, +0.79], cluster-bootstrapped by task) and +0.80 on the eight- and nine-family matrices, where the larger cell count tightens the estimate. The most anti-aligned family pairs transfer at or below chance, and the single best-aligned pair on the four-open matrix (Gemma and Llama, $\cos +0.29$) is also its best-transferring off-diagonal pair. The relationship survives the action-only canonicalization (rho +0.66, CI [+0.51, +0.78]), so the misalignment is behavioral, not a stylistic artifact of family prose. The off-lineage monitor fails not because it is miscalibrated but because it ranks suspicion along the wrong direction, one fit to a lineage whose sabotage looks geometrically different. This is the concept shift of Section 5 made concrete, and it is why benign-only recalibration, which can only translate or rescale rather than rotate the head, cannot recover the loss.

The relationship is not an artifact of attacker obviousness or monitor capability. Residualizing both the cosine and the transfer AUROC on an additive per-calibration-family and per-evaluation-family model, the same main-effect subtraction that defines our interaction term, and correlating the residuals gives a partial Spearman rho of +0.78 on the four-open matrix (95% CI [+0.63, +0.89]) and +0.83 and +0.79 on the eight- and nine-family matrices, at or above the raw correlations. Head geometry predicts the calibrate-on-A, test-on-B transfer structure net of the same main effects our headline removes, so the misaligned-direction mechanism explains the interaction itself, not merely which families are easy to flag. It attenuates but survives the action-only canonicalization (partial rho +0.35, CI [+0.15, +0.52]).

The relationship predicts held-out cells rather than redescribing fitted ones. In a leave-one-family-out cross-validation, a two-parameter line fit on the off-diagonal cells of all other lineages predicts a held-out lineage’s transfer AUROC with an out-of-sample Spearman of +0.78 (eight families) and +0.79 (nine families), each clear of zero and essentially equal to the in-sample correlation, and cuts the held-out prediction error by more than half against a predict-the-mean baseline (MAE skill +0.54 and +0.59). Leave-one-cell-out agrees to within 0.004, so the signal does not come from sibling cells sharing a family with the held-out one. On the four-open primary matrix the cross-validated estimate is too small to resolve (twelve off-diagonal cells; the in-sample rho there is suggestive), so the predictive claim rests on the larger rosters. The cosine requires labeled sabotage from both families, so this is a post-hoc explanation rather than a deployable benign-only predictor, consistent with the Section 5 prescription that the transfer matrix cannot be skipped.

The angle is causal. A reviewer can ask whether head misalignment drives the off-lineage collapse or only tracks it, so we intervene on the angle directly. Because the strict score is affine in the embedding, the transfer AUROC depends only on the head’s direction, so we rotate calibration head A toward evaluation family B along the geodesic between their directions and read the AUROC at each angle. As the head rotates onto B the transfer AUROC recovers monotonically (within-pair Spearman of AUROC against rotation fraction +0.95 on the nine-family matrix and +0.89 on the four-open matrix), and rotating halfway recovers 86% of the diagonal-to-off-diagonal gap on the nine-family matrix (95% CI [+0.84, +0.88], cluster-bootstrapped by task) and 84% on the four-open matrix (CI [+0.78, +0.90]). The recovery is specific to the direction toward B: an equal-magnitude rotation in a random direction, averaged over 64 azimuths, recovers nothing (+0.00 [-0.02, +0.02] on four-open, -0.01 [-0.02, -0.01] on nine families), so the gain is not an artifact of perturbing the head or adding parameters, and the aligned-minus-random contrast clears zero on every roster including action-only. It is also unrecoverable from benign data, which is why the per-lineage recalibration of Section 5 cannot be skipped: a mean-shift of B’s benign cloud onto A’s leaves the AUROC ranking exactly unchanged because a linear head is translation-invariant, and CORAL alignment of the benign covariances actively hurts, its recovery -0.24 of the gap (CI [-0.30, -0.20] on nine families), the concept-shift control of Section 6 made constructive. Finally the rotation can be estimated from a small labeled set: a difference-of-means direction probe from m labeled sabotage and m benign transcripts from B recovers about half the gap by m around 10 and roughly 80% by a near-full per-family set, tracing the

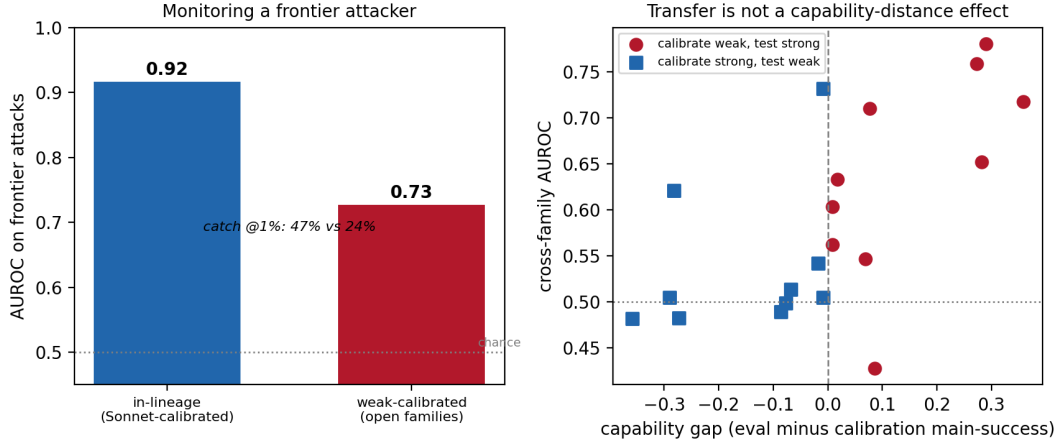


Figure 10: Capability gap: weaker-model calibration roughly halves frontier detection (left); transfer is not a capability-distance effect (right).

same cosine-to-AUROC law as the oracle rotation, while a label-shuffled probe of equal size recovers nothing. The off-lineage monitor fails because its discriminative direction is rotated away from the new family’s sabotage, and the only thing that rights it is labeled sabotage from that family, the measurement the Section 5 prescription says cannot be skipped.

5 A protocol for cross-family monitoring

The transfer gap is a deployment cost: under the defer-to-trusted protocol of Section 4.4, at a fixed 1% audit budget an off-lineage attacker is roughly twice as likely to slip past auditing as the in-lineage attacker whose detection a monitor’s published number reports. A control-eval designer or a deployer who calibrates a trusted monitor on one lineage and faces another should run the four-step protocol below. Each step is a result we measured and pre-registered on the four open families rather than an exhortation, and together they pre-empt the natural objections that one could re-threshold, triage, or pool the gap away.

Why a monitor overfits its lineage (mechanism). Benign transcripts are strongly family-separable. A four-way classifier on honest embeddings recovers the producing family at 2.6x chance (0.656 accuracy). The per-family attack head keys on this stylistic signature, which is why it ranks in-family attacks well and foreign attacks poorly. Stripping prose and embedding only the submitted code (action-only) reduces the APPS interaction to +0.055, about half the full-transcript value, but leaves it clearly positive with CI excluding zero, so the overfit is part stylistic and part a real behavioral component. On bigcodebench the action-only interaction is +0.099 on the leak-free basis, a behavioral component comparable to APPS.

5.1 Step 1: Report the full cross-family transfer matrix, never a single pairing

The first question is whether the drop can be predicted cheaply, from benign transcripts alone, before any sabotage from the new family is seen. It cannot. Benign distributional distance between the calibration and target family (a cross-validated two-sample classifier in the embedding space) does not predict the per-pair AUROC drop out of family: a genuine leave-one-family-out fit, trained on pairs not involving the held-out family, predicts its drop level at Spearman near zero, and the only in-sample-significant association (action space, seven families, Mantel $p=0.030$) fails to generalize. Family-specific main effects dominate the drop and do not track benign separability, so the gap cannot be inferred from family resemblance. The only valid number is therefore the measured one: a control-eval paper should publish the calibrate-on-A, test-on-B grid and label which cells are in-lineage, which

costs a paper that already generates multi-family transcripts nothing beyond the scoring it has run.

5.2 Step 2: Detect off-lineage operation at inference from benign drift

A deployer can at least detect at inference that traffic has gone off-lineage. The squared Mahalanobis distance to the calibration family’s benign cloud separates in-lineage from off-lineage benign transcripts at mean AUROC 0.836 (full) and 0.681 (action-only) over off-diagonal pairs, both above 0.65. The trigger to recalibrate exists even though the size of the loss does not. The detector predicts neither: run through the identical leave-one-family-out test it also fails to predict the transfer-drop level (pooled LOFO -0.06 on the primary roster), so detecting off-lineage operation and sizing its cost are separate capabilities of the same signal.

5.3 Step 3: Recalibrate per lineage with labeled sabotage; do not input-patch or benign-align

The cheap patches fail, which is what forces recalibration. A per-input out-of-distribution signal cannot triage which inputs to defer to a trusted model and recover the lost detection, with no labels from the new family: deferring the most out-of-distribution inputs does not beat spending the same deferral budget on the monitor’s own suspicion ranking (the difference is within the bootstrap CI of zero at 5, 10, and 20% budgets in both spaces) and is consistently worse than random deferral. Off-lineage sabotage is not more out-of-distribution than off-lineage benign (the OOD score’s own attack-versus-benign AUROC on the new family is 0.55, near chance), so the deferred inputs are mostly off-lineage benign traffic. Off-lineage operation can be detected, but not patched input by input.

Benign data alone does not suffice either. Aligning the new family’s benign cloud onto the calibration family’s, with the attack head frozen and no new sabotage labels, recovers none of the gap: a pure mean shift leaves AUROC exactly unchanged (a constant translation under a linear head shifts every score equally and cannot change the ranking, which also makes concrete that the Track-2 ranking gap is not an operating-point artifact), and second-order CORAL alignment recovers under 2% of the gap and is negative on the four-family setting (-0.008 full, -0.011 action-only, CI excluding zero). The failure is a concept shift, not a covariate shift: the calibration head’s discriminative direction is simply wrong for the new family’s sabotage, because the sabotage signature itself is family-specific. Recalibration therefore requires labeled sabotage from the faced lineage, and the learning curve is sublinear. 150 labeled attacks plus 150 benign per off-diagonal cell recover only about 81% of the gap (full) and 58% (action-only), and reaching 90% needs roughly 300, essentially a full per-family sabotage set per cell. The off-lineage monitor is a useful warm start only at tiny labeling budgets (at 5 to 20 labeled examples, upweighted augmentation beats a from-scratch head by 0.006 to 0.023 AUROC with CI excluding zero), and it becomes a liability past 150 examples, where pooling its wrong-direction labels drags the head off the new family’s optimum. The constructive conclusion is sharp: a deployer can tell they are off-lineage but cannot cheaply close the gap, so a control evaluation cannot substitute family resemblance or benign-traffic recalibration for a measured cross-family transfer matrix built from real sabotage in each lineage.

5.4 Step 4: Treat multi-family pooling as a real but strictly bounded partial defense

The most natural remaining defense is to calibrate the monitor on a panel of families rather than one. At the four-family setting this looked hopeless: pooling the other three families’ calibration data gave no defensive payoff for a held-out family and could anti-transfer. At eight families, where the roster is finally large enough to draw a curve, the answer is a real but strictly bounded partial defense. Sweeping the number of pooled calibration families from one to seven, with each held-out target family scored by a head trained on a panel it never faces and a target-stratified cluster bootstrap, pooling more families significantly beats single-family calibration: the matrix-comparable gain at seven pooled families is +0.031 ([+0.023, +0.040], CI excluding zero), in the leave-one-family-out direction the framework

predicts. But the curve plateaus by about three families and recovers only about a fifth of the in-distribution detection gap; even pooling all seven foreign lineages leaves the monitor at 0.685 AUROC against an own-family ceiling of 0.825, so most of the gap is irreducible by pooling. The recovered fraction is also stylistic, not behavioral: in action-only space the pooling benefit is small under a shared embedding basis (+0.024) and vanishes under a strictly held-out basis (flat at chance), so a diverse panel recovers some family-style transfer but none of the family-specific behavioral signal. A diverse calibration panel is therefore a cheap, statistically significant, but bounded and style-only mitigation, consistent with the concept-shift diagnosis above. The family-adaptive alternatives turn out to be roster-dependent. On the four-family setting, routing a held-out family to its nearest calibration family by benign-centroid cosine recovered cross-family AUROC from 0.539 to 0.616; on the five-family setting that adds one frontier model the payoff vanished (routing 0.55 and foreign-head ensembling 0.60 versus a 0.59 single-foreign baseline), because the frontier family had no good neighbor in the pool. On the full six-family setting with two frontier vendors, the recovery returns and is significant on average: routing reaches 0.657 (cluster-bootstrap [0.645, 0.669]) and benign-z-scored head ensembling 0.648, against a single-foreign baseline of 0.625 ([0.616, 0.634]). The average is carried by strong families pairing up (the two frontier-adjacent lineages route to each other at gains of +0.14 to +0.22), and per-family reliability is still not guaranteed: routing hurts three of six held-out families. The diagnosis also holds at scale with one correction: projecting out the top family-identity directions trims the six-family interaction from +0.137 to +0.110 by rank 3, but the effect is non-monotone (back to +0.134 by rank 10 while off-diagonal AUROC falls from 0.625 to 0.572), so identity-subspace removal degrades everyone’s discriminability without eliminating the diagonal advantage; the removable-style share is small at scale. A family-invariant pooled head (project out the family-identity subspace, then train one attack head on the pooled remaining families) ties rather than beats the single-foreign baseline on the five-family setting (0.587 versus 0.588). We therefore frame the constructive contribution as follows: with a diverse enough calibration roster, family-adaptive scoring (routing or ensembling over per-family heads) recovers a real fraction of the transfer gap on average, but no tested method is reliable per-family, so the primary recommendation remains to report cross-family transfer matrices and to calibrate per-family wherever the deployed lineage is known; a robustly general family-adaptive defense is open.

6 Confound controls

The interaction survives the controls that distinguish calibration overfit from the two confounds. Table 5 summarizes them; each row re-estimates the interaction under a different robustness check, and all exclude zero, though the magnitude varies: matching attacks for blatancy, the strongest alternative explanation, reduces it most. Unless noted, controls are run at the four-family scale ($n=200$) where they were established; the headline scale-up and frontier results are in Section 4.

- **Difficulty matching.** The four open families differ in attack main-task success by about 14 points. Subsampling each family’s test attacks to an identical main-success composition and re-scoring the same monitors leaves the interaction at +0.123 [+0.102, +0.147], stable across subsample seeds. At the five-family scope the frontier model widens the capability spread to 36 points (Qwen 0.32 to Sonnet 0.68); matching all five to a common composition (155 attacks per family at 53.5% main-success) leaves the interaction at +0.135 [+0.123, +0.147], essentially identical to the unmatched +0.132. The diagonal advantage is not a difficulty artifact, even with a frontier family in the matrix.
- **Blatancy matching (the dominant confound).** The strongest alternative to calibration overfit is that a family’s sabotage is simply more or less incriminating, not that the monitor keys on its style. Using the fixed-prompt monitor’s per-transcript suspicion as a blatancy proxy, we subsample each family’s test attacks to an identical suspicion-band composition (mean detectability equalized across families) and re-score. On the leak-free basis the full-transcript interaction is +0.133 [+0.113, +0.154] under proxy #1 (the fixed-prompt suspicion band), down only modestly

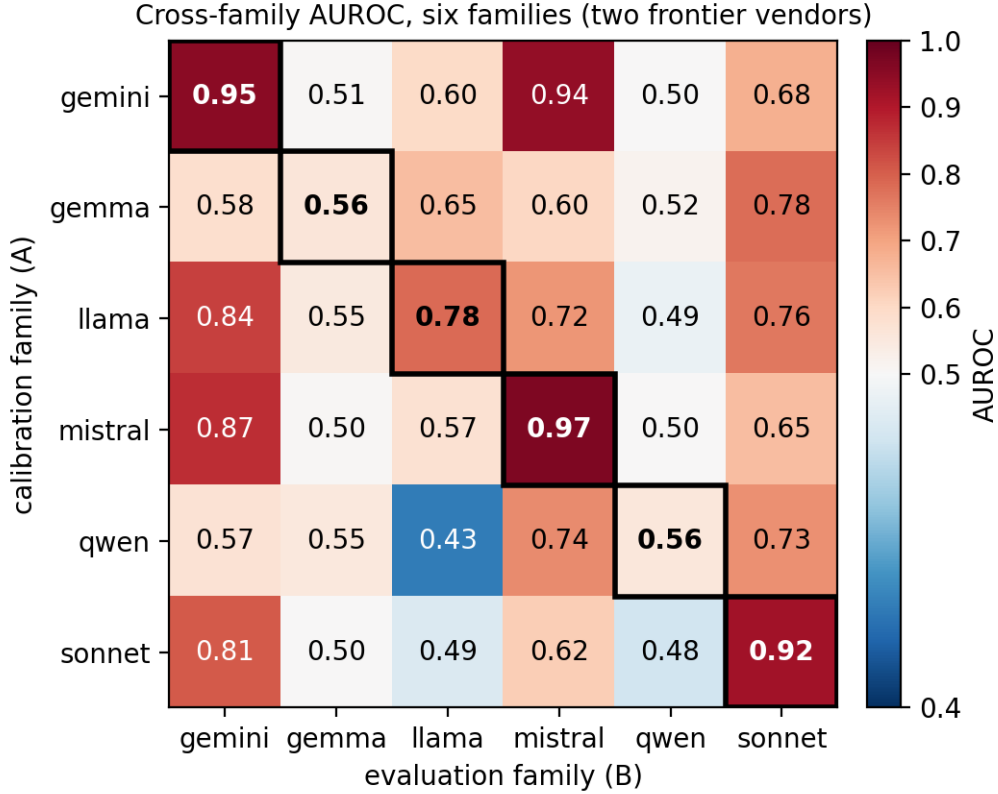


Figure 11: Six-family cross-family AUROC (two independent frontier vendors, Sonnet and Gemini).

from the $+0.172$ headline; the transductive basis read this residual far smaller, $+0.045$ [$+0.025$, $+0.064$], because the pooled preprocessing diluted it, and that number is the conservative lower bound. A second, monitor-independent proxy, a held-out Haiku blatancy grade, gives an even larger residual of $+0.176$ [$+0.162$, $+0.190$], so the residual does not depend on a monitor-coupled blatancy measure. The residual is significant and stable across five subsample seeds, and under the transductive proxy it reappears within both the most-subtle and most-blant suspicion strata taken alone ($+0.048$ and $+0.045$). Blatancy thus carries much of the raw diagonal bonus, consistent with the incrimination-dominance literature, but a real family-specific residual persists at matched detectability (families equalized on the fixed-prompt suspicion band and, independently, on the held-out Haiku grade, the two blatancy proxies above); this is what backs the claim that a monitor ranks its own lineage’s equally detectable sabotage higher, where “equally detectable” means matched on those proxies rather than absolutely. The behavioral component also survives the joint control on the leak-free basis: stripping family prose to action-only and equalizing blatancy at once leaves the interaction at $+0.046$ [$+0.024$, $+0.069$] (the transductive basis read $+0.017$ [$+0.000$, $+0.031$], where the leak suppressed it), stable across five subsample seeds. It concentrates in the subtle-attack stratum ($+0.075$ [$+0.056$, $+0.094$]) and is null on blatant attacks, so the residual is a behavioral transfer failure on exactly the sabotage that detection is for. Most of the jointly-controlled gap is still style-carried (the full-transcript estimate is $+0.133$), but a behaviorally real, equally detectable own-family advantage remains.

- Embedder specificity, within and across vendors. Re-embedding with text-embedding-3-large ($+0.122$) and ada-002 ($+0.109$) reproduces the interaction; the Mistral diagonal stays near 0.96 and the Mistral-head-to-Llama anti-transfer stays below chance in all three. This is not an OpenAI-embedder artifact: re-embedding

Control	Variant	Interaction	95% CI
Baseline	logistic head, text-embedding-3-small	+0.127	[+0.112, +0.143]
Difficulty matching	matched	+0.123	[+0.102, +0.147]
	main-success (4 families)		
Difficulty matching	matched	+0.135	[+0.123, +0.147]
	main-success (5 families, 36-pt spread)		
Blatancy matching	matched suspicion band (4 families, n=600)	+0.045	[+0.025, +0.064]
Embedder	text-embedding-3-large	+0.122	[+0.106, +0.138]
Embedder	text-embedding-ada-002	+0.109	[+0.095, +0.125]
Head class	MLP	+0.111	[+0.097, +0.126]
Head class	SVM (RBF)	+0.159	[+0.141, +0.177]
Head class	random forest	+0.128	[+0.108, +0.149]
Head class	gradient boosting	+0.115	[+0.097, +0.134]
Regularization	PCA truncated to 10 components	+0.061	[+0.043, +0.080]

Table 5: The interaction under each confound control, with the strict leak-free re-estimation reported as primary in Table 5b. Every estimate is positive with a 95% CI excluding zero; intervals are the multiplicity-preserving task-cluster bootstrap. The transductive magnitudes in this table are the conservative lower bound, larger on the leak-free basis in every case.

Control	Strict (leak-free, primary)
Difficulty matching (4 families)	+0.159 [+0.138, +0.183]
Blatancy matching, proxy #1 (fixed-prompt suspicion band)	+0.133 [+0.113, +0.154]
Blatancy matching, proxy #2 (held-out Haiku grade)	+0.176 [+0.162, +0.190]
Head-overfit ablation (PCA-50 baseline)	+0.158 [+0.139, +0.178]
Head-overfit ablation (PCA-10, brutal cut)	+0.149 [+0.122, +0.177]
Head class (five heads)	+0.130 to +0.163

Table 5b: Strict leak-free re-estimation of the confound controls, reported as primary; the Table 5 transductive values are the conservative lower bound. The strict basis fits the scaler, PCA, and head on the calibration family alone, with the evaluated family absent from the pipeline; the control design is otherwise identical to Table 5. Every control is larger leak-free and every CI excludes zero.

the four-family n=600 transcripts with a non-OpenAI, open-weight embedder (nomic-embed-text-v1.5, a different architecture and vendor) gives interaction +0.097 (95% CI [+0.086, +0.108]) full and +0.043 ([+0.033, +0.054]) action-only, both within ~ 0.01 of the OpenAI text-embedding-3-small values (+0.110, +0.055) at the same scale. The embedding monitor’s overfit is thus a property of family-conditioned embedding in general, not of one embedder vendor.

- Head-class. Linear, neural, kernel, and tree heads all give CI-positive interactions (Section 4.2).
- Head-overfit regularization. The interaction is invariant to L2 strength across three orders of magnitude and survives a brutal cut to 10 PCA components (+0.061), the opposite of what small-n head overfit would do.

7 Limitations

- The mean effect is carried by rankable lineages. Mistral and Llama have high own-family diagonals; the weakest attackers (Qwen, Gemma) sit near chance on their own diagonal because they barely produce detectable backdoors. The interaction is net of main effects and bootstrapped, so it accounts for this, but the claim is that off-lineage calibration collapses detection for the lineages a monitor can rank, not uniformly across all families.
- Calibration, not scale. The main matrix is at 600 transcripts per family; the interaction holds (+0.110) with a roughly one-third tighter interval and the realized-safety gap firms to +6.5 points. A residual calibration finding remains: the fixed-prompt monitor supports a 1% operating point only for Mistral; for the other three families the benign score distribution saturates at the score ceiling, so a 1% operating point does not exist for them, a sharper form of the claim that the operating point is family-specific.
- Family count and serving. We nominate the frozen four-open-family matrix as the pre-registered primary result and label every later family addition (five through nine families, and the frontier-only sub-matrices) as post-hoc robustness. Nine families on APPS, five of them frontier-capability models spanning five independent vendors (Anthropic Claude Sonnet, Google Gemini, xAI Grok, OpenAI GPT-5, DeepSeek; Section 4.5), three on bigcodebench (Gemma’s fragile multi-turn submission is deferred). Open families on DeepInfra, the frontier families on their native APIs. The capability-gap axis (calibrate-weak, test-strong) is addressed by reanalysis in Section 4.6; a dedicated multi-tier capability design that varies monitor and attacker capability jointly remains planned.
- Scope: trained, family-conditioned monitors. The interaction is established for trained monitors (frozen-embedding heads, LoRA sequence classifiers up to 34B, and a closed-API fine-tuned frontier monitor; Section 4.2). The prompted monitors a deployer would run did not clear the competence gate on both gated families: across six constructions (four few-shot, one rubric-distilled, one retrieval-augmented over two judges; Section 4.2) every prompt-only monitor ranks Llama’s attacks at or below chance, even when handed per-query retrieval of in-family references whose own kNN signal clears the gate (Llama 0.729) and reproduces the interaction (+0.130). The family signal is present in context but a prompted judge reverts to absolute red-flag judgment and cannot exploit it; trained calibration is required, and once it is trained the gap reappears: the per-family gemini-2.5-flash fine-tunes clear the gate on both gated families (Mistral 0.738, Llama 0.706, the latter narrowly) and reproduce the interaction at +0.100 (95% CI [+0.063, +0.134]). The tuned weights are closed and not releasable, so the open ladder remains the reproducible backbone. We keep the broad framing in the title but scope the established claim to trained, family-conditioned monitors and report the prompted-monitor shortfall, with its mechanism, as a characterized limitation.
- Monitor robustness. The LoRA interaction is stable across three training seeds (+0.179, +0.204, +0.194) and two base models at fixed small scale (Phi-3.5-mini +0.179, Phi-4-mini +0.180, both 3.8B; the capability staircase in Section 4.2 climbs to 14B and 34B separately), all CIs excluding zero; the residual caveat is that both bases share a vendor (Microsoft), though both are vendor-disjoint from all four policy families.
- Success framing. We report unfiltered honest-versus-attack AUROC, which measures detection of attack attempts, because both-success backdoors are near zero at this scale for sub-frontier policies (5 of 8 families have 0 to 1 successful-backdoor test positives), so a realized-sabotage matrix is not computable post-hoc. Restricting instead to attacks that completed the main task, a competent-attacker slice that is computable, leaves the leak-free interaction at +0.169 (four open families) and +0.201 (eight families), indistinguishable from the unfiltered +0.172 and +0.197, so the ranking advantage is not an artifact of failed attacks (Section 4.1). On the four frontier lineages that do realize sabotage we can go further: GPT-5 reaches 50 both-success positives, Grok reaches 41 with its APPS task pool fully exhausted, and the interaction survives as a non-saturating effect size of +0.738 ([+0.407, +1.027]),

about 1.1 times the attempt-level effect (AUROC is ceiling-bound at +0.030 here; Section 3). Only GPT-5 clears the preregistered 50-positive gate, so this is a robustness slice rather than a powered realized-sabotage matrix; powering the remaining cells (DeepSeek at 15, Sonnet at 7) is the main open item, gated on generating a substrate with a higher clean-backdoor yield.

8 Related work

Our closest neighbor is CoT Red-Handed (Arnav et al., 2025), which sweeps multiple monitors and attacker families and self-calibrates each monitor per untrusted model at 1% FPR. That is a diagonal-only evaluation; it never holds the calibration family fixed and measures the off-diagonal. Our wedge is the calibrate-on-A, test-on-B matrix net of attacker obviousness. Information-Access (Arike et al., 2026) is the closest precedent in setup: it fixes a monitor and varies attacker families on BigCodeBench-Sabotage with the same gpt-4.1-mini monitor we use, and is the source of our TPR-at-1%-FPR reporting and success-filtering conventions. Its axis is the monitor’s information access, how much of the agent’s trajectory the monitor reads, where it reports a less-is-more effect and an extract-and-evaluate remedy, not the calibration family; it never builds the calibrate-on-A, test-on-B matrix or decomposes the cross-family gap into attacker obviousness, monitor capability, and the interaction, which is our contribution. Control Tax (Terekhov et al., 2025a) reports an attacker-by-monitor AUROC matrix on the same APPS setting but uses raw AUROC with no fixed-A/test-B calibration and no interaction decomposition, naming calibration as future work. Async Control (Cooper Stickland et al., 2025) fixes a calibrated monitor and tests out-of-distribution, but shifts the task with the policy held fixed; our axis is the policy family. Reliable Weak-to-Strong Monitoring (Kale et al., 2025) has a capability-gap axis but reports raw AUROC and TPR rather than the obviousness-netted interaction; our Section 4.6 shows the cross-family failure is not a capability-distance effect, separating the two. We differentiate from adaptive-attack work (Terekhov et al., 2025b) and attack-selection work (Schaeffer et al., 2026b) explicitly: we study passive distribution shift with no monitor targeting, and that work’s finding that realized safety is more sensitive to false-positive rate than to true-positive rate is consistent with our budget-enforced protocol holding the false-positive rate fixed across families. The closest result in shape is Cross-LLM Generalization of Behavioral Backdoor Detection (Sanna, 2025), a calibrate-on-A, test-on-B matrix across six LLMs whose within-distribution detection accuracy of 92.7% falls to 49.2% off-model, a 43-point cross-LLM drop recoverable by adding model identity as a feature. That raw drop is the quantity our decomposition exists to interrogate: a cross-model gap of that size could be attacker obviousness (off-model backdoors are simply harder to spot), detector capability, or calibration overfit, and raw accuracy with a model-identity feature cannot separate the three. Our reported effect is the share that survives netting the first two out as main effects, with a task-clustered interval. The domain differs too: their object is a trained supply-chain malware classifier, not a trusted sabotage monitor inside a control protocol with an audit budget, so the cross-model drop carries no realized-safety reading. Directionally their collapse corroborates ours; isolating the calibration-overfit component from attacker obviousness is the gap we close. Practical Challenges of Control Monitoring (Lindner et al., 2025) catalogs deployment-scale monitoring difficulties such as latency and incremental attacks, but its axis is monitoring latency rather than transfer across policy families. Ensemble Monitoring for AI Control (Koran et al., 2026) shows diverse monitor ensembles beat homogeneous ones out of distribution, which varies the monitor side at a fixed attacker rather than holding the calibration family fixed. Evaluating Control Protocols for Untrusted AI Agents (Kutasov et al., 2025a) stress-tests defer and resample protocols on SHADE-Arena under adaptive red-teaming, an adaptive-attack axis distinct from our passive cross-family shift. A fresh literature sweep (2026-06-13) found no paper building the calibrate-on-A / test-on-B transfer matrix with an interaction decomposition for sabotage monitors. The general worry that reported accuracies overstate deployment safety is partly raised elsewhere: classifier monitors lose accuracy as the transcript grows (Martin & Roger, 2026), an external-validity axis orthogonal to the policy family we hold fixed, and a control-relevant policy property such as intervention awareness is itself model-specific and must be re-measured per release rather than assumed invariant (Schaeffer et al., 2026a).

The cross-family calibration-validity mechanism and its realized-safety translation are the unmade contribution.

9 Conclusion

Trusted sabotage monitors are not automatically valid across untrusted policy families. The own-family advantage survives as a positive interaction after removing attacker obviousness and monitor capability, replicates across monitor classes and substrates (and strengthens when frontier attackers spanning five independent vendors are added), and translates into a substantial loss in realized catch rate at a fixed audit budget, including roughly halved detection when a weaker-model-calibrated monitor is deployed against the frontier. Control evaluations should report cross-family transfer matrices, because a single-pairing accuracy is a diagonal cell and the deployed monitor lives off the diagonal, and should calibrate per-family wherever the deployed lineage is known; naive multi-family pooling and single family-adaptive heuristics do not reliably close the gap, which we leave open.

A Reproducibility

Code: `scripts/track2_embed.py` (embeddings), `track2_matrix.py` (matrix + interaction + cluster bootstrap), `track2_difficulty.py` (difficulty-matched), `track2_headclass.py`, `track2_ablation.py`, `track2_invariant.py` (mitigation), `track2_lora_monitor.py` (LoRA), `track2_safety.py` (realized safety). Split: md5-seed-0, 60/40 task-disjoint. Provenance (open-weight HF revisions, frontier API aliases with generation dates, ControlArena SHA) in `configs/provenance.yaml`; closed-API frontier models expose only a routing alias rather than a pinnable weight revision, which we record as such. Generation funnels and per-cell matrices in `docs/CONTROLS_RESULTS.md`, `SAFETY_TRANSLATION.md`, and `BIGCODEBENCH_RESULT.md`. Raw transcripts are gated under the benchmark canary strings; analysis code and aggregate results are released.

A.1 Roster registry

All rosters, no omissions. The four-open matrix is the frozen pre-registered primary; every later roster is post-hoc robustness with its inclusion rule stated below, each pre-registered (prereg and result docs dated 2026-06-10 to 2026-06-15).

Roster	Families	Role	Inclusion rule
4-family open	Qwen, Mistral, Gemma, Llama	frozen primary	Pre-registered; all four generated before scoring, none dropped; the interaction excludes zero here before any frontier family is added
5-family	+ Claude Sonnet	post-hoc robustness	One frontier vendor, vendor-disjoint from the monitor
6-family	+ Gemini	post-hoc robustness	Second independent frontier vendor
8-family	+ Grok, GPT-5	post-hoc robustness	Two more independent vendors
9-family	+ DeepSeek	post-hoc robustness	Fifth frontier vendor
frontier-only	the frontier subset	post-hoc robustness	Sharpest transfer test, every attacker a strong model

Roster	Families	Role	Inclusion rule
bigcodebench	Mistral, Llama, Qwen	second code substrate	Same families, different code distribution (a second code-backdoor setting, not a second safety domain)
SHADE	Sonnet, GPT-4.1, Gemini	non-code substrate	Confirmatory only, reported as underpowered

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